SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY COURSE CODE: CSA5115

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

*I **Nissanki Sri Venkata Lakshmi Kavya (192373036)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

N.S.V.L.Kavya

Annexure A
Scenario-based

1. Secure Data Transmission using Block Cipher
A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

A. **Explain why this issue occurs.**
B. **Suggest a more secure block cipher mode and justify your choice based on the scenario.**

1. Secure Data Transmission using Block Cipher

A. Explain why this issue occurs.

Answer:

In Electronic Codebook (ECB) mode, each plaintext block is encrypted independently using the same key.

- If two plaintext blocks are identical, they produce the same ciphertext block.
- Since the company's financial data contains repeated blocks, the ciphertext also contains repeated patterns.
- An attacker can observe these patterns and gain information about the data, making ECB less secure.

Example:

Plaintext : A A B A

ECB Output: X X Y X

Here, identical plaintext blocks (A) produce identical ciphertext blocks (X), revealing patterns.

B. Suggest a more secure block cipher mode and justify your choice.

Answer:

The company should use Cipher Block Chaining (CBC) mode.

Justification:

- In CBC mode, each plaintext block is XORed with the previous ciphertext block before encryption.
- The first block uses an Initialization Vector (IV).
- Even if the plaintext blocks are identical, the ciphertext blocks will be different because each block depends on the previous ciphertext.
- This hides patterns in the data and provides better security than ECB.

Example:

Plaintext : A A B A

CBC Output: X Z Y W

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

A. Identify the most suitable algorithm for this scenario.

B. Justify your choice based on security strength and efficiency.

A. Identify the Most Suitable Algorithm

Answer:

The **3-DES (Triple DES)** algorithm is the most suitable choice for a banking application because it provides **high security**, which is the primary requirement for protecting customer transactions.

B. Justify Your Choice Based on Security Strength and Efficiency

Algorithm	Security	Performance
DES	Low (56-bit key, vulnerable to attacks)	Fast
3-DES	High (168-bit key, much stronger than DES)	Moderate
Blowfish	High (up to 448-bit key)	Fast

Justification:

- **DES** is not suitable because its **56-bit key** is too small and can be broken using modern computing power.
- **3-DES** applies the DES algorithm **three times** with a **168-bit key**, providing much stronger security. Although it is slower than DES and Blowfish, it offers the high level of security required for banking transactions.
- **Blowfish** is faster and secure, but **3-DES has been widely used in banking systems** and is trusted for protecting financial transactions where security is more important than speed.

Conclusion:

Since the banking application requires **high security with moderate performance**, **3-DES** is the most appropriate choice because it provides stronger encryption than DES while offering acceptable performance for secure customer transactions.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand